



Skole

How the system works

A short, plain-language guide: who uses the school system, what each person can do, how the data fits together and how it runs.

Live site: <https://skoleph.web.app>

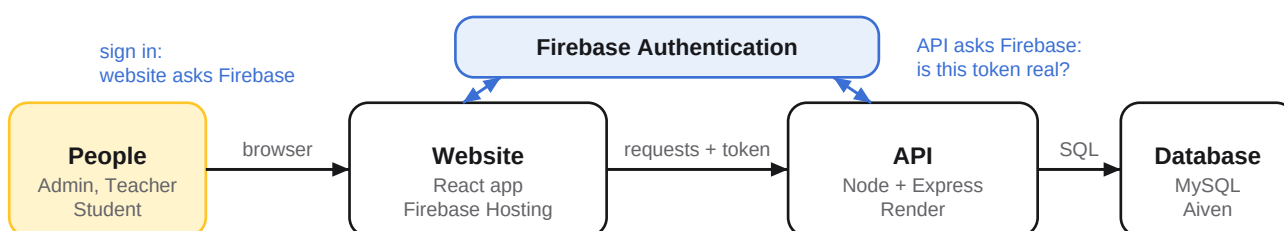
Source code: <https://github.com/Lianhahaha/School-Management-System>

Demo sign-in: admin@school.test, teacher1@school.test or student1@school.test, password Password123!

1. The big picture

Skole is a web application for running a school. It works in any browser, with nothing to install. Administrators, teachers and students sign in, and each sees only what belongs to them.

Piece	Job	Runs on
Website	The screens you click (React).	Firebase Hosting
API	The back office: checks who you are, applies the rules, reads and writes data.	Render
Database	Stores every school record.	Aiven MySQL
Sign-in	Knows email and password and proves who you are. Holds no school data.	Firebase Authentication



Every school record lives in MySQL. Firebase only knows who you are (email + password).

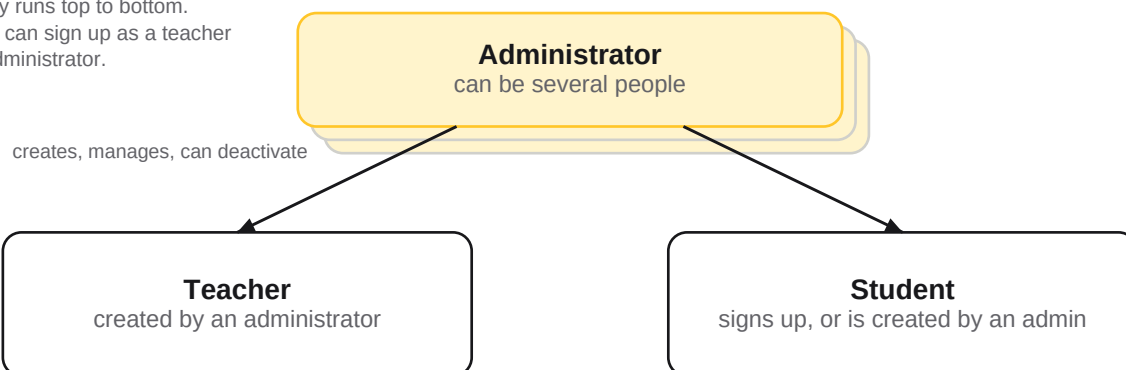
Why split it this way

The website never touches the database. Every action goes through the API, which checks who you are and what you may do first, so editing a web address cannot reveal anyone else's data.

2. People and authority

There are three roles. A role is chosen when the account is created and never changes.

Authority runs top to bottom.
Nobody can sign up as a teacher or an administrator.



Administrator: the highest authority. Manages all accounts and all school data. **There can be several** (the demo has two). Nobody can sign up as one: an existing administrator creates it, and the first comes from the setup command.

Teacher: works with the classes they are assigned to: marks attendance, creates assessments and enters grades for their own lessons, posts announcements to their classes.

Student: sees their own class, timetable, attendance, grades and announcements. Read-only, except their own contact details.

The administrator is protected from mistakes

You cannot deactivate or delete **your own** account. The **last active administrator** can never be removed, even if two try at once. An account with school records can be deactivated but not deleted. Role and email cannot be edited after creation.

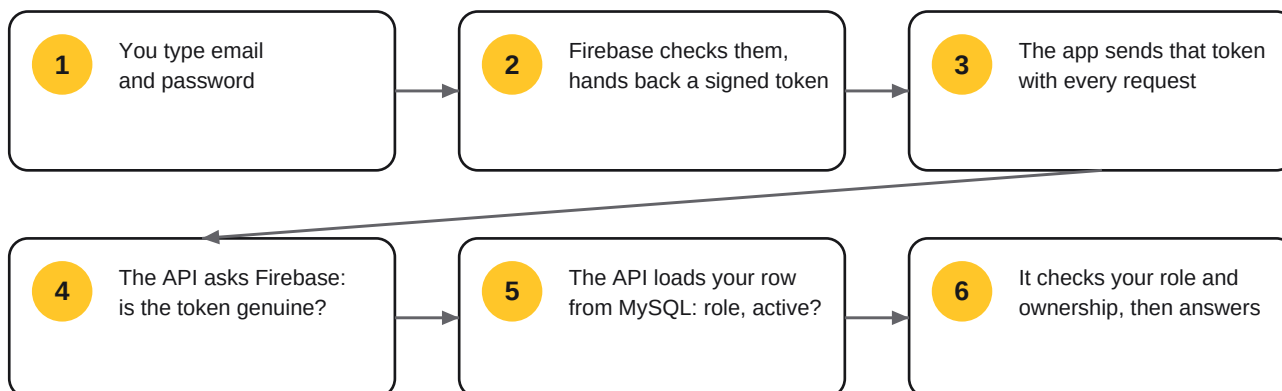
Capability	Admin	Teacher	Student
Create and deactivate accounts	Yes	No	No
Subjects, classes, teacher assignment, enrollments, timetable: change	Yes	No	No
Student profiles: view	All	Their classes	Themselves
Attendance: mark	Yes	Own lessons	No
Assessments and grades: enter	Yes	Own lessons	No
Attendance and grades: view	All	Own lessons	Own record
Announcements: post	Anyone	To own classes	No
Edit own contact details	Yes	Yes	Yes

3. Accounts and sign-in

Role	How the account is made
Administrator	Created by an administrator (Users, Create user). The first one comes from the setup command.
Teacher	Created by an administrator, who sets a temporary password and gives it to the teacher. Teachers cannot sign up themselves.
Student	Signs up on the public Create account page, or an administrator creates it. With no class yet, only the profile shows until an administrator enrolls them.

There is no "promote" button. A role decides which records an account owns, so it is fixed. A teacher who signed up as a student by mistake is deleted (while unused) and recreated as a teacher.

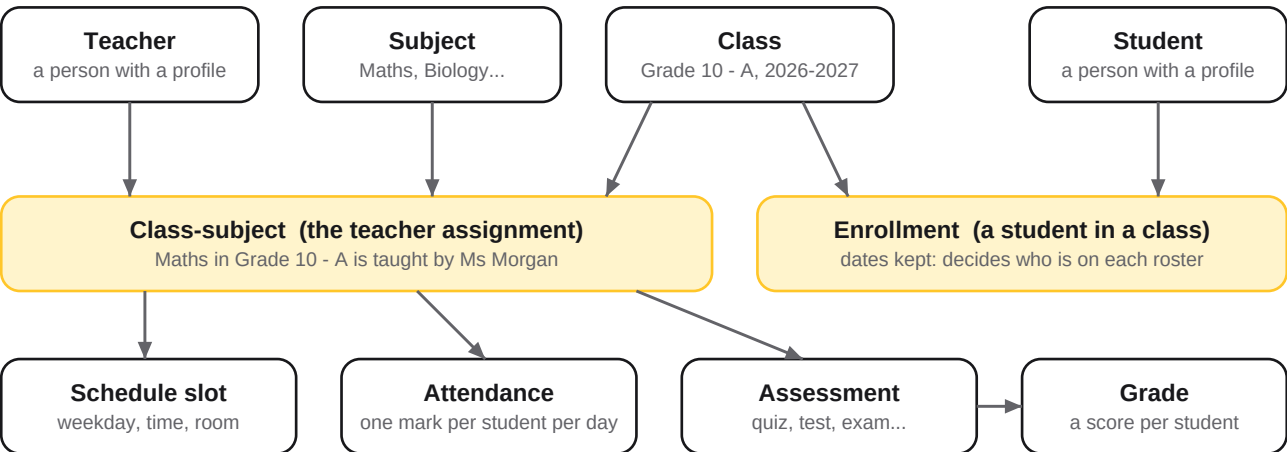
What happens when you sign in



This runs on **every request**, so a deactivated person is locked out on their very next click.

- **Forgot password** emails a reset **link** (not a code). The same message appears whether or not the email exists.
- Demo accounts use fake addresses and cannot receive email. To try it, register a student with a real address.
- **Deactivate** blocks access and keeps all records (a student's active enrollment is withdrawn). **Delete** is only for an unused, mistaken account.

4. How a school is organised



Concept	What it is, and the rules
Subject and class	A subject is taught (Maths). A class is a group for one school year, with a grade level and an optional homeroom teacher. The year starts in August (2026-2027). Anything in use cannot be deleted.
Class-subject	A subject in a class taught by one teacher. It anchors the timetable, attendance and grades. Changing the teacher keeps earlier records.
Enrollment	A student in a class, with dates. One active enrollment at a time. A transfer closes the old one and opens a new one, so history is never overwritten.
Timetable	Weekly slots (day, time, room). Refused if the class, teacher or room is already booked; the error lists the clashing lessons.
Attendance	Per lesson and day: present, absent, late or excused. The roster is the students who were in the class that day. No future dates. Rate = (present + late) / marked.
Assessments and grades	A quiz, test or exam with a maximum score. A score must be 0 to the maximum. Results are points-weighted (total scored / total possible), so bigger tests count more.
Announcements	Audience (everyone, teachers or students), optional class, optional publish and expiry dates. Teachers post only to their own classes and edit only their own posts.

5. A day in the life

Administrator	Teacher	Student
1. Create teachers and students 2. Create subjects and classes 3. Assign a teacher to each subject 4. Enroll students 5. Build the timetable 6. Watch the dashboard, handle changes	1. Read the Today dashboard 2. Mark attendance 3. Create an assessment, enter the scores 4. Post an announcement to a class 5. Check the weekly timetable	1. See today's timetable 2. Check attendance rate 3. Read grades per subject 4. Read announcements

Dashboards differ by role. The administrator sees counts, attendance today and upcoming assessments; the teacher sees today's periods and pending grading; the student sees their timetable, attendance and latest grades. Every list has search, filters, sorting and paging, and a light or dark theme switch. On phones the menu becomes a bottom bar.

6. Safety and rules

- Every request is checked for a real sign-in, an active account and the right role. The API enforces it even if a screen is bypassed.
- Input is validated strictly, SQL is parameterised, and only the allowed websites may call the API from a browser.
- The database itself enforces key rules (unique emails, one active enrollment, valid scores), so a code bug cannot break them.
- A refused action shows a clear message: 401 not signed in, 403 not allowed, 404 not found, 409 conflict (duplicate, in use, clash), 400 invalid input.

7. Hosting, data and limits

What	Where	Good to know
Website	skoleph.web.app (Firebase Hosting)	Free, always on
API	skole-api.onrender.com (Render)	Free; sleeps after 15 idle minutes, the next request takes about a minute
Database	Aiven MySQL free plan	Free, 1 GB; may be switched off after long inactivity (turn it back on in Aiven)

Where the data lives. Nothing is stored on anyone's computer, so changing laptops loses nothing. The browser keeps only the signed-in session and the theme choice.

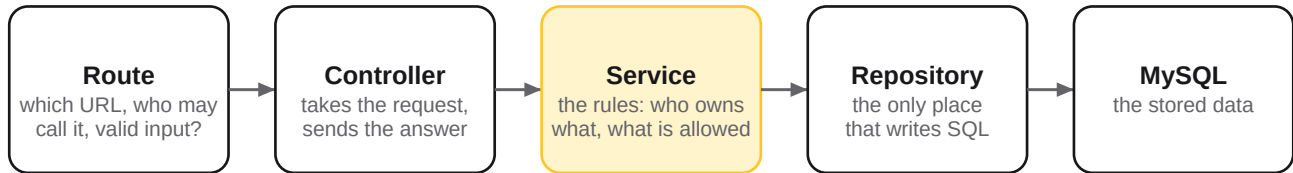
Real time? Everyone shares the same data and every save is written at once. An open page **refreshes itself about every 20 seconds** while the tab is visible (and when you return to it), so other people's changes appear without reloading. The attendance and grade sheets a teacher is filling in are never refreshed under their hands. If two people edit the same record, the last save wins.

Run it locally: the README lists every step (Node 24, MySQL 8, a free Firebase project, then `npm run db:migrate`, `npm run db:seed` and `npm run dev` in backend and frontend).

Demo accounts (password Password123!): admin@school.test and admin2@school.test, teacher1 to teacher3, student1 to student8, all @school.test.

8. Under the hood

React 19 and Tailwind on the website, Node.js 24 and Express 5 for the API, MySQL 8 with 12 tables, Firebase Authentication, 64 documented API endpoints and 128 automated tests. Every feature follows the same path:



The **service** holds the rules and the **repository** is the only place that writes SQL, so a bug is found by following one path. The website hides what you cannot do, but the API is what enforces it.

9. Quick answers

Question	Answer
Can there be only one administrator?	No, as many as needed. At least one active administrator always remains.
Is the administrator above everyone?	Yes. They manage accounts and all school data.
How does a teacher get an account?	An administrator creates it and gives the teacher a temporary password.
Can a student become a teacher?	No. Roles are fixed. Delete the unused account and create a new one.
What if a student changes class?	The administrator transfers them. The old enrollment stays as history.
Does deactivating erase data?	No. Records stay. The person simply cannot sign in until reactivated.
Why was the first page slow?	The free server sleeps when idle and wakes on the first request.